Banking Customer Churn Prediction Analysis Summary

Project Overview

Objective: Predict customer churn using Logistic Regression on banking customer data
Dataset Size: 10,000 customers with 12 features
Model: Logistic Regression (binary classification)
Test Split: 80% training, 20% testing (2,000 test samples)

Dataset Insights

1. Dataset Composition

- **Total Customers:** 10,000
- **Features:** 11 (after removing customer_id)
- **Target Variable:** Churn (1 = Churned, 0 = Not Churned)
- **Data Quality:** No missing values detected

2. Feature Categories

Numerical Features:

- credit_score: Customer creditworthiness (range: ~350-850)
- age: Customer age
- tenure: Years with the bank (0-10 years)
- balance: Account balance
- products_number: Number of products held (1-4)
- estimated_salary: Estimated annual income

Binary Features:

- credit_card: Has credit card (0/1)
- active_member: Active membership status (0/1)

Categorical Features (Encoded):

- gender: Male/Female → One-hot encoded to gender_Male
- country: France/Germany/Spain → One-hot encoded to country_Germany, country_Spain

3. Data Preprocessing Steps

Removed: customer_id (non-predictive identifier)
Encoding: One-hot encoding applied to gender and country
Scaling: StandardScaler applied to normalize all features (mean=0, std=1)
Train-Test Split: 80-20 split with random_state=42 for reproducibility

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Model Performance Metrics

Overall Accuracy: 81.1%

The model correctly predicts churn status for 81.1% of customers in the test set.

Confusion Matrix Breakdown

		Predicted			
		Not Churn		Churn	
Actual	Not Churn	1543		64	True Negatives | False Positives
	Churn	314		79	False Negatives | True Positives

Interpretation:

- **True Negatives (1,543):** Correctly predicted customers who didn't churn
- **True Positives (79):** Correctly predicted customers who churned
- **False Positives (64):** Incorrectly predicted churn (but customer stayed)
- **False Negatives (314):** Missed churners (predicted stay but actually churned)

Detailed Performance Analysis

Class 0 (Not Churned Customers)

- **Precision:** 83% - When model predicts "not churn", it's correct 83% of the time
- **Recall:** 96% - Model identifies 96% of all non-churning customers
- **F1-Score:** 0.89 - Excellent balanced performance for this class
- **Support:** 1,607 customers (80.4% of test set)

Insight: The model is very strong at identifying customers who will stay with the bank.

Class 1 (Churned Customers)

- **Precision:** 55% - When model predicts "churn", it's correct 55% of the time
- **Recall:** 20% - Model only identifies 20% of all churning customers
- **F1-Score:** 0.29 - Poor performance indicating significant room for improvement
- **Support:** 393 customers (19.6% of test set)

Critical Issue: The model struggles significantly with identifying actual churners, missing 80% of them (314 out of 393).

Key Findings & Insights

Strengths

1. **High Overall Accuracy (81.1%):** Good general performance
2. **Excellent Non-Churn Detection:** 96% recall for class 0 means very few false alarms about potential churners
3. **Low False Positive Rate:** Only 64 customers incorrectly flagged as churners (4% of non-churners)
4. **Clean Data Pipeline:** Proper preprocessing with scaling and encoding

' **Critical Weaknesses**

1. **Poor Churn Detection (20% recall):** The model misses 4 out of 5 actual churners
    - **Business Impact:** 314 customers who churned were not identified, representing significant lost revenue opportunities
2. **Class Imbalance Issue:**
    - Non-churners: 80.4% of dataset
    - Churners: 19.6% of dataset
    - The model is biased toward the majority class (not churn)
3. **Low Churn Precision (55%):** Nearly half of churn predictions are false alarms
    - **Business Impact:** Wasted resources on retention efforts for customers who weren't going to leave

Conclusion

Summary

The Logistic Regression model achieves 81% accuracy but suffers from severe class imbalance issues. While it excellently identifies customers who won't churn (96% recall), it catastrophically fails at detecting actual churners (only 20% recall).

Primary Issue

The model is too conservative - it defaults to predicting "no churn" for most customers, leading to:

- Very few false alarms (good for resource efficiency)
- Missing 80% of actual churners (terrible for business retention goals)